
A Methodology to Locate and Search Submarines

Summary

Sightseeing submarines are very popular in today's tourism industry, but the series of disappearances caused by mechanical failures also urgently need attention and resolution.

In this paper, we have identified three strategies to help MCMS Company complete the route prediction of the distressed submarine, the allocation of optimal rescue equipment, and the determination of the fastest rescue plan.

To predict the possible drift route of the distressed submarine, we have established the **Submarine Position Prediction Model**. Based on the modeling of the seabed terrain features, we have determined the **Ocean Current Generation Model**. Combined with the analysis of the ocean density jump layer, we finally determined the five-degree-of-freedom kinematic equation and dynamic equation of the submarine. We combined the above models to simulate the drift trajectory of the distressed submarine and obtained the drift prediction image of the distressed submarine.

Next, in order to select the most beneficial rescue equipment, we have established the **Cov-AHP Search Equipment Evaluation Model**. This model is an optimization based on the traditional **Analytic Hierarchy Process (AHP)**, which avoids the subjective influence brought by expert evaluation and improves the credibility of the model.

In the rescue of the distressed submarine, we established **The Fastest Submarine Rescue Strategy Model**. We first used the Submarine Random Walk Algorithm and Monte Carlo Method to obtain the submarine detection probability distribution. Based on this probability distribution, we determined the important path points in the rescue, formed a weighted undirected connected graph, and thus converted the submarine rescue problem into a Traveling Salesman Problem. Then we used the Genetic Algorithm to simulate the model, thereby deriving the relationship between the cumulative distribution function (c.d.f.) of the found submarine and time.

Finally, we extended the above models to the situations of different sea areas and multiple submarines, demonstrating that the models have a wide range of applicability and good stability.

According to the results of the model operation, we found that the distressed submarine drifted a horizontal distance of 35 to 40 kilometers and a depth of about 2 kilometers after losing power for 15 hours. We recommend that MCMS Company equip the main ship with Side-Scan Sonar as the main rescue equipment. We predict that the probability of finding the submarine is 91.52% only 27 minutes after dispatching the rescue ship.

We also provided a report memorandum to the Greek government, explaining the detailed content and practical value of our entire rescue plan.

Keywords: Five-degree-freedom dynamics, Cov-AHP, Monte Carlo Method, Genetic Algorithm

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1 Introduction

1.1 Background

Taking a sightseeing submarine for sightseeing is an interesting and adventurous activity. Passengers can reach the bottom of the sea and appreciate some unusual scenery by taking a sightseeing submarine. However, this activity has both opportunities and risks. When a mechanical failure occurs, the situation can become very dangerous. To enhance the safety of submarine sightseeing, we can provide guiding suggestions for the safe operation and rescue of submarines through mathematical modeling methods.

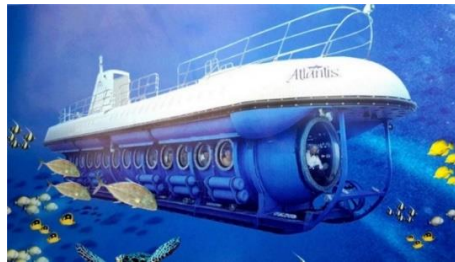


Figure 1-1: A picture of a typical sightseeing submarine

1.2 Restatement of the Problem

Based on our understanding of submarines and the information we have found, combined with the requirements of the problems, the question can be restated as follows:

- **Problem 1**
 - Considering various interference factors, establish a model to **predict the position of the submarine**.
 - Based on the established model, evaluate what **information** the submarine should regularly send to the main ship.
- **Problem 2**
 - By investigating and evaluating the advantages and disadvantages of various search devices, decide **which devices should be installed on the main ship**.
 - Evaluate what **other equipment** should be carried on the **rescue ship**.
- **Problem 3**
 - Develop a model to evaluate the best **deployment method for search equipment**.
 - Analyze the **probability** of finding the submarine.
- **Problem 4**
 - Apply the model to the situation with **multiple submarines**.
 - Apply the model to **different sea areas**.

1.3 Our Work

To establish a location prediction model suitable for submarines, we conducted the following work:

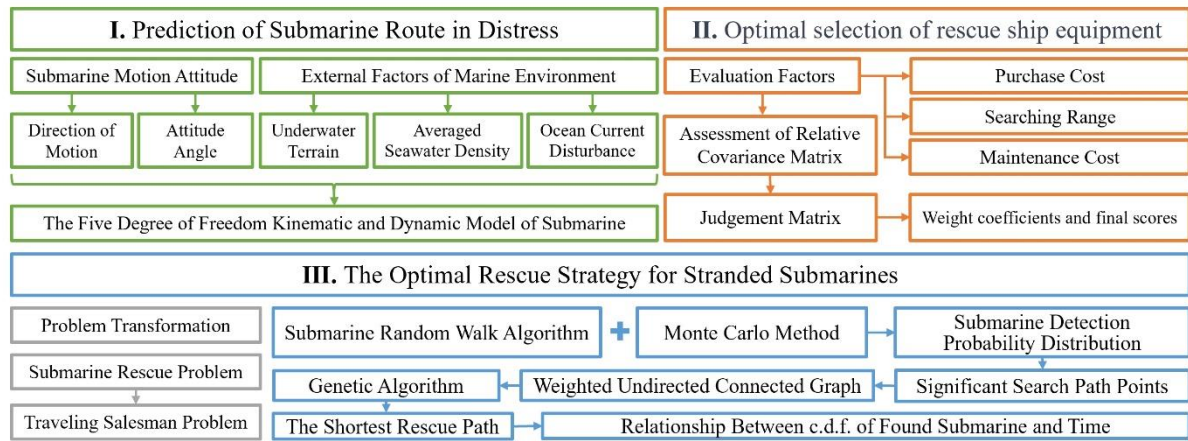


Figure 1-2: Our work

2 Assumptions and Justifications

- **Assumption 1:** The direction of the velocity of the ocean currents remains constant over a short period.
 - **Justification:** Ocean current is a large-scale geographical phenomenon, where their velocity and direction can be considered as constant over short periods and smaller spatial scales.
- **Assumption 2:** The rescue boat has a certain search range, and once the submarine enters this range, it can definitely be found.
 - **Justification:** Doing this can effectively simplify the model and avoid the occurrence of too many uncertain factors.

3 Notations

The primary notations used in this paper are listed in this following table.

Notations		
Symbol	Description	Unit
ρ	The density of seawater	kg/m^3
V_{cd}	The velocity of ocean current	m/s
ψ	The heading angle	degree
θ	The pitch angle	degree
φ	The roll angle	degree

4 Task 1: Submarine Position Prediction

To predict the submarine position, these following models are established:

4.1 Underwater Terrain Model

To simulate the terrains under the sea, the complex geographical environment under water can be abstracted as following: uphill, downhill, convex, concave^[1].

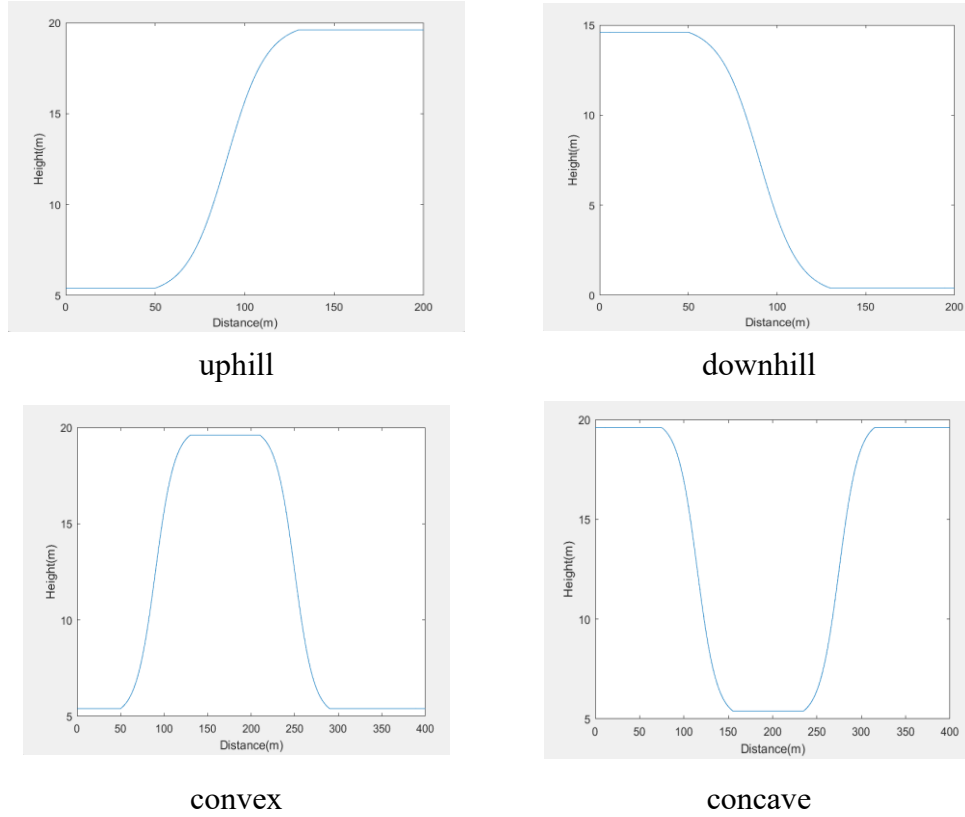


Figure 4-1: Four different terrain patterns

In the following modeling, based on these four typical terrain patterns, we will analyze other characteristics of ocean currents.

4.2 Ocean Density Profile Model

The characteristic of seawater density stratification is a factor that cannot be ignored in the ocean^[2]. When the seawater density stratification is strong, submarines may encounter difficulties and dangers when they want to sink or float, which may have a significant impact on the navigation and safety of the submarine^[3].

Based on the stratification characteristics of seawater density, the existing models often use the hyperbolic tangent function to simulate the ocean density profile model^[4]. In addition, we have also introduced two parameters, the density stratification depth and the density stratification thickness, to describe the characteristics of the density stratification, making the model more widely applicable and able to simulate more complex hydrological environments. The model can be expressed as this following formula:

$$\rho(h) = \bar{\rho} - \frac{\Delta\rho}{2} \tanh\left(\frac{h - h_{str}}{d_{str}}\right) \quad (1)$$

In this formula, $\bar{\rho}$ represents the density of sea water, $\Delta\rho$ is the difference of density

stratification. h_{str} is the depth of density stratification, d_{str} stands for the thickness of density stratification.

When the average density $\rho = 1025.25 \text{ kg/m}^3$, the stratification density difference $\Delta\rho = 6.5 \text{ kg/m}^3$, the stratification depth $h_{str} = 10\text{m}$, the stratification thickness $d_{str} = 200\text{m}$, and when the stratification depth $h_{str} = 300\text{m}$, the stratification thickness $d_{str} = 350\text{m}$, the density-depth image is simulated through Python as shown in the figure.

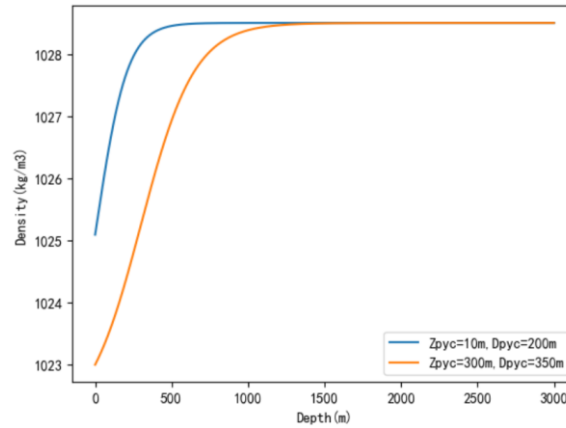


Figure 4-2: The density-depth graph

The curve finally simulated is basically consistent with the average ocean density curve provided in [4] in summer, which proves the effectiveness of our model.

By using the characteristic parameters of seawater density distribution to analytically describe the seawater density distribution, it can provide convenience for the measurement of ocean density at the location of the submarine, thereby providing the possibility for the establishment of the submarine's dynamics equation.

4.3 Ocean Current Generation Model

In this section, we focus on modeling how ocean currents are generated in the ocean.

The magnitude of ocean currents in the real world is highly intricate, influenced by various environmental factors such as wind speed, temperature, and Seabed topography. Their speeds rang from just a few centimeters per second in most locations to 1–2 meters per second in specific areas [5].

In the absence of detailed hydrological features of the sea area, most scholars treat ocean currents as steady and uniform flows when studying the mathematical model of underwater vehicle motion. In other words, the size and direction of the ocean current do not change with the change of time and spatial point position. The main factors affecting the ocean current are wind speed, sea water depth, and sea surface slope [6].

For submarine search and rescue, the submarine is at a great depth, and the ocean current is less affected by wind speed [5]. Therefore, this section focuses on the impact of seawater depth and seabed topography on ocean currents. The speed of the ocean current changes continuously with depth, and the relationship between its speed and depth can be approximately described by a parabolic function[6]:

$$V_{cd} = 2.5v \times \ln(6.34 \times 10^6 v^* h) \quad (2)$$

Which h stands for the height of seawater from the bottom of the sea, $v^* = \sqrt{ghs}$, s is the slope of the surface of seawater.

By combining the modeling of the seabed topography in the previous section, we can easily calculate the sea level at a specific location in the ocean, thereby obtaining the speed of the ocean current.

4.4 Five Degrees of Freedom Dynamics Model

Underactuated Model - Establishment and Transformation of Coordinate Axes

In order to accurately analyze the motion state of the submarine, we first need to establish coordinate systems for the ocean environment and the submarine itself. Here, we use the inertial coordinate system and the body coordinate system [7,8].

The inertial coordinate system and the body coordinate system are orthogonal coordinate systems as shown in **Figure 4-3**. The three positive directions of the inertial coordinate system point to the north, east, and the center of the earth respectively, and its coordinate origin E is located at a certain point in the sea. The origin O of the body coordinate system is located at the center of gravity of the submarine, and Ox , Oy , Oz point to the bow, starboard, and bottom of the submarine respectively.

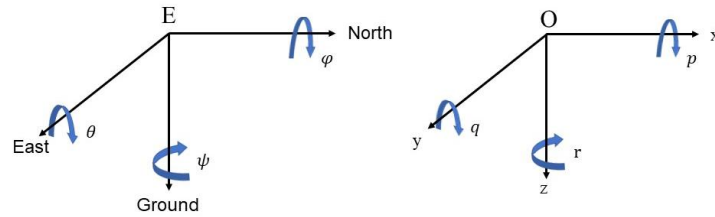


Figure 4-3: The inertial coordinate system and the body coordinate system

In this case, we assume that the origins of the two coordinate systems coincide, and the way they transform each other is only by rotating the roll angle φ around the Ox axis, the pitch angle θ around the Oy axis, and the heading angle ψ around the Oz axis to make the two coordinates coincide. We represent the position and angle of the submarine in the inertial coordinate system as $[x \ y \ z]^T$ and $[\varphi \ \theta \ \psi]^T$, respectively, and the velocity and angular velocity in the body coordinate system as $[u \ v \ w]^T$ and $[p \ q \ r]^T$, respectively.

Formula (3) shows the relationship between the velocity of submarine between the angle velocity:

$$\left[\frac{dx}{dt} \ \frac{dy}{dt} \ \frac{dz}{dt} \right]^T = J_1 \cdot [u \ v \ w]^T \quad (3)$$

Where J_1 is a relationship matrix, its specific form is as follows:

$$J_1 = \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta\sin\varphi - \sin\psi\cos\varphi & \cos\psi\sin\theta\cos\varphi + \sin\psi\sin\varphi \\ \sin\psi\cos\theta & \sin\psi\sin\theta\sin\varphi + \cos\psi\cos\varphi & \sin\psi\sin\theta\cos\varphi - \cos\psi\sin\varphi \\ -\sin\theta & \cos\theta\sin\varphi & \cos\theta\cos\varphi \end{bmatrix} \quad (4)$$

Formula (5) shows the relationship between the angle and angular velocity of the submarine.

$$\left[\frac{d\varphi}{dt} \frac{d\theta}{dt} \frac{d\psi}{dt}\right]^T = J_2 \cdot [p \ q \ r]^T \quad (5)$$

Similarly, J_2 is also a relationship matrix:

$$J_2 = \begin{bmatrix} 1 & \sin\varphi \tan\theta & \cos\varphi \tan\theta \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \frac{\sin\varphi}{\cos\theta} & \frac{\cos\varphi}{\cos\theta} \end{bmatrix} \quad (6)$$

In the above formula, the attitude angles φ , θ , ψ should respectively satisfy the following constraints:

$$\begin{cases} \varphi \in (-\pi, \pi) \\ \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \\ \psi \in (-\pi, \pi) \end{cases} \quad (7)$$

Also, the relationship matrix should also satisfy $J_1^{-1} = J_1^T$ and $J_2^{-1} = J_2^T$.

Underactuated Model - Five Degrees of Freedom Equation

In order to simplify the model of the submarine and make it conform to the underactuated fault state, we reduce it from the original six degrees of freedom to five degrees of freedom, and combine fluid mechanics and other related knowledge to represent the kinematic model and dynamic model of the submarine with five degrees of freedom. The following set of equations (8) shows the five-degree-of-freedom kinematic equation, and the set of equations (9) shows the five-degree-of-freedom dynamic model:

$$\begin{cases} \frac{dx}{dt} = u \cos\psi \cos\theta - v \sin\psi + w \cos\psi \sin\theta \\ \frac{dy}{dt} = u \sin\psi \cos\theta + v \cos\psi + w \sin\psi \sin\theta \\ \frac{dz}{dt} = -u \sin\theta + w \cos\theta \\ \frac{d\theta}{dt} = q \\ \frac{d\psi}{dt} = \frac{r}{\cos\theta} \end{cases} \quad (8)$$

$$\begin{cases} \frac{du}{dt} = \frac{m_1}{m_2} \cdot vr - \frac{m_3}{m_1} \cdot wq - \frac{d_1}{m_1} \cdot u \\ \frac{dv}{dt} = -\frac{m_1}{m_2} \cdot ur - \frac{d_2}{m_2} \cdot v \\ \frac{dw}{dt} = \frac{m_1}{m_3} \cdot uq - \frac{d_3}{m_3} \cdot w + \frac{g_1}{m_3} \\ \frac{dq}{dt} = \frac{m_3 - m_1}{m_4} \cdot uw - \frac{d_4}{m_4} \cdot q - \frac{g_2}{m_4} + \frac{\tau_q}{m_4} \\ \frac{dr}{dt} = \frac{m_1 - m_2}{m_5} \cdot uv - \frac{d_5}{m_5} \cdot r + \frac{\tau_r}{m_5} \end{cases} \quad (9)$$

The specific meanings of some parameters in these two sets of equations are as follows:

$$\begin{aligned}
& \text{added masses} \begin{cases} m_1 = m - X_{\dot{u}} \\ m_2 = m - Y_{\dot{v}} \\ m_3 = m - Z_{\dot{w}} \\ m_4 = I_y - M_{\dot{q}} \\ m_5 = I_z - N_{\dot{r}} \end{cases} \\
& \text{the influence of gravity and buoyancy} \begin{cases} g_1 = (W - B)\cos\theta \\ g_2 = (Z_g \cdot W - Z_b \cdot B) \end{cases} \\
& \text{damp coefficient} \begin{cases} d_1 = X_u + X_{u|u}|u| \\ d_2 = Y_v + Y_{v|v}|v| \\ d_3 = Z_w + Z_{w|w}|w| \\ d_4 = M_q + M_{q|q}|q| \\ d_5 = N_r + N_{r|r}|r| \end{cases}
\end{aligned} \tag{10}$$

Where m represents the mass of the submarine, I_y, I_z represent the moment of inertia of the submarine around the Oy axis and Oz axis respectively, $X_{(\cdot)}, Y_{(\cdot)}, Z_{(\cdot)}, M_{(\cdot)}, N_{(\cdot)}$ are a series of added mass coefficients and viscous hydrodynamic coefficients, W, B represent the size of the gravity and buoyancy received by the submarine respectively, Z_g, Z_b are the coordinates of the buoyancy center and center of gravity on the vertical axis of the submarine in the body coordinate system, τ_q is the roll torque, τ_r is the yaw torque. According to the literature^[9], we have obtained specific data for some parameters, shown in the **Appendix 1**.

Underactuated Model - Ocean Current Disturbance

The above models are all established under ideal conditions. In fact, when the submarine loses power, the impact of ocean currents on its movement is not to be ignored. Therefore, we need to establish a submarine motion model under the disturbance of ocean currents.

We assume that the ocean current is constant and let its speed in the body coordinate system be $U_c = [u_c, v_c]^T$, where u_c and v_c represent the longitudinal and transverse speeds of the ocean current, respectively. Then let the speed of the submarine relative to the ocean current be $U_x = [u_x, v_x]^T$, similarly, u_x and v_x represent the relative longitudinal speed and relative transverse speed, respectively. Therefore, the actual speed of the submarine can be represented as $U = [u_c + u_x, v_c + v_x]^T$. Combined with relevant kinematics knowledge, we can obtain the submarine motion equation under the disturbance of ocean currents:

$$\begin{cases} \frac{dx}{dt} = (u_x + u_c)\cos\psi - (v_x + v_c)\sin\psi \\ \frac{dy}{dt} = (u_x + u_c)\sin\psi + (v_x + v_c)\cos\psi \\ \frac{d\psi}{dt} = r \end{cases} \tag{11}$$

4.5 Result of Submarine Position Prediction

Based on the seabed depth simulated by the seabed topography modeling, we can obtain the image of the ocean current speed in the ocean. Due to the complex and changeable hydrological conditions of the ocean, to simulate the complex ocean current environment more realistically in the ocean, we have added random disturbances based on the formula provided by

the previous ocean current generation model. When the disturbance coefficients are 0.3, 0.35, and 0.4 respectively, we can obtain the following three ocean current images:

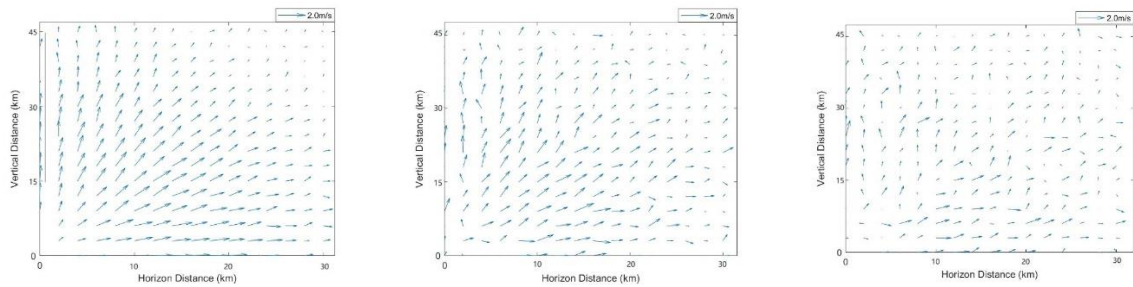


Figure 4-4: Ocean current patterns under different disturbances

From these images, we can see that when the seabed depth distribution and seawater density are consistent, the overall direction of the ocean currents is the same. However, turbulence will occur in a small part of the sea area. If a submarine sails through these areas, it may be subject to strong disturbances. The maximum ocean currents in the three images are 1.623m/s , 1.479m/s , and 1.515m/s , respectively.

By incorporating these three ocean current scenarios into the five-degree-of-freedom submarine mechanical model, we use MATLAB to simulate the relationship between the submarine's position and time. It can be expressed as this following image:

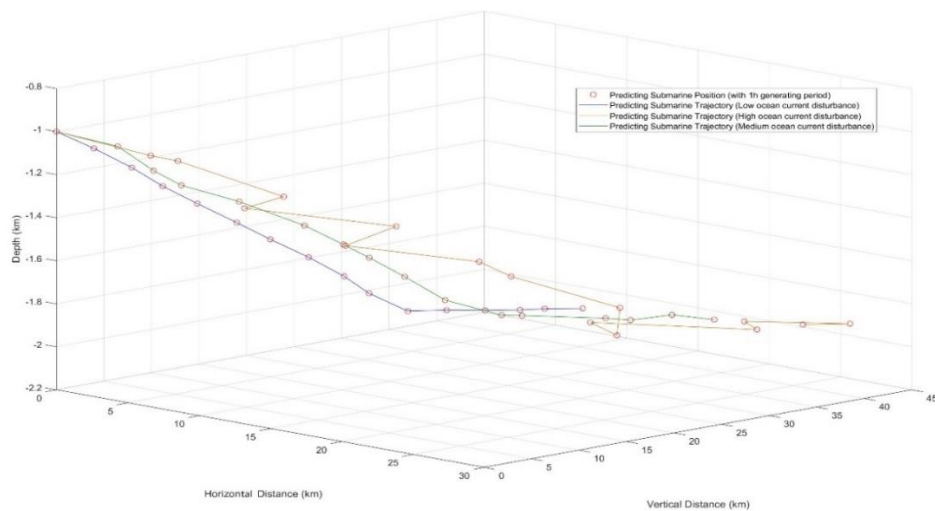


Figure 4-5: Graph of submarine position changing over time

This image simulates the position of the submarine during the fifteen hours after it lost contact. The three lines correspond to three different ocean current scenarios, and the points in the image represent the submarine's position every hour. When a submarine loses power and its buoyancy system fails, it will drift with the ocean current. As we can see on **Figure 4-5**, 15 hours after losing contact, the relative coordinates of the submarine are $(16.39\text{km}, 30.69\text{km}, -2.12\text{km})$, $(19.97\text{km}, 32.26\text{km}, -2.05\text{km})$, and $(24.42\text{km}, 32.52\text{km}, -1.94\text{km})$ respec-

tively. In general, the planar drifting distance of the submarine after 15 hours of losing contact is between 35km and 40km, and it finally floats at a depth of about 2km from the sea level.

This section uses the previously proposed seabed topography model and ocean current generation model to obtain the ocean current image under specific hydrological conditions. In addition, with the help of the ocean thermocline density model and the five-degree-of-freedom submarine model, we obtained the time-position image of the lost contact submarine under this ocean current condition. This provides a feasible reference for submarine search and rescue.

4.6 What information should be sent?

Based on the results obtained from the aforementioned model, we believe that the submarine should regularly transmit the following data to the main ship: **current position, speed and heading, depth, and ocean current velocity**. With the help of this regularly updated information, we can timely grasp the current movement and trend of the submarine, providing useful references for possible rescues.

5 Task 2: Evaluation of the Equipment that Should be Installed

To evaluate what should be installed on the main ship, we established the following model:

5.1 Cov-AHP Search Equipment Evaluation Model

When a submarine encounters an accident, maritime search and rescue equipment becomes a crucial aid for the main vessel to locate the missing submarine. To select the most suitable maritime search and rescue equipment, in this section, we improved upon the Analytic Hierarchy Process (AHP) and established the Cov-AHP Search and Rescue Equipment Evaluation Model to identify the most efficient maritime search and rescue equipment.

The traditional AHP faces a challenge, as it requires several experts to make subjective judgments about the relative importance of various elements within the system based on their own experiences and knowledge ^[10]. In order to mitigate the subjective influence of expert ratings, we have adopted the evaluation matrix construction method from [11] — the Cov-AHP approach.

The steps to obtain the evaluation matrix are as follows:

Step 1: Based on the quantitative indicator data reflected by the elements, calculate the covariance matrix A . Use a_{ij} to represent the covariance in the i -th row and j -th column.

<i>item</i>	x_1	x_2	...	x_n
x_1	a_{11}	a_{12}	...	a_{1n}
x_2	a_{21}	a_{12}	...	a_{2n}
...	...	...	...	...
x_n	a_{n1}	a_{n2}	...	a_{nn}

Step 2: Transform the covariance matrix A into the relative covariance matrix B . Use the following formula to transform the covariance matrix into the relative covariance matrix.

$$\begin{cases} \frac{b_{ij}}{b_{ii}} & i \neq j \\ 1 & i = j \end{cases} \quad (12)$$

At this point, the relative covariance matrix B should have the following form:

item	x_1	x_2	...	x_n
x_1	1	b_{12}	...	b_{1n}
x_2	b_{21}	1	...	b_{2n}
...	...	...	...	...
x_n	b_{n1}	b_{n2}	...	1

Step 3: Construct the judgment matrix. To determine if a matrix is a judgment matrix, two fundamental properties must be checked. Firstly, all values on the diagonal must be 1. Secondly, each value should be the reciprocal of its corresponding value^[10]. Clearly, for the matrix B at this stage, it is not yet a judgment matrix. Therefore, it is necessary to transform matrix B so that the product of all corresponding elements' values equals 1. The transformation method is as following:

$$c_{ij} = \frac{b_{ij}}{\sqrt{b_{ij} \times b_{ji}}} \quad (13)$$

The matrix C , composed of c_{ij} , is the evaluation matrix.

item	x_1	x_2	...	x_n
x_1	1	c_{12}	...	c_{1n}
x_2	c_{21}	1	...	c_{2n}
...	...	...	...	...
x_n	c_{n1}	c_{n2}	...	1

After obtaining the judgment matrix, the next step is to proceed with the consistency test for the judgment matrix.

First, we obtain the maximum eigenvalue λ_{max} of the n -order judgment matrix through eigen decomposition. Then, according to the formula (14), we calculate the Consistency Index (CI) of the judgment matrix.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (14)$$

After that, based on the order n of the judgment matrix, find the corresponding Random Index (RI) from the **Table 5-1**.

Table 5-1: Random Index (RI)^[11]

n	1	2	3	4	5	6	7	8	9
RI	0	0	0.52	0.89	1.12	1.26	1.36	1.41	1.46

Finally, by calculating the ratio of CI to RI , we can obtain the Consistency Ratio (CR) of the judgment matrix.

$$CR = \frac{CI}{RI} \quad (15)$$

At this point, if $CR < 0.1$, it indicates that the consistency of the judgment matrix is acceptable, meaning that the judgment matrix is relatively reasonable.

After confirming the reasonableness of the judgment matrix, we can proceed to calculate the weights of various column factors in the judgment matrix. Assuming a column in the judgment matrix is represented as $[b_1 \ b_2 \ \dots \ b_n]^T$, the weight for that column is

$[\frac{b_1}{\sum_{i=1}^n b_i} \ \frac{b_2}{\sum_{i=1}^n b_i} \ \dots \ \frac{b_n}{\sum_{i=1}^n b_i}]^T$. After calculating the weights for each column in this manner, we obtain a weight matrix with the same order.

To simplify the computation, we take the arithmetic average of the weights in each row, resulting in the comprehensive weights for each system factor.

5.2 Result of Cov-AHP Search Equipment Evaluation Model

After conducting our research, we found that we can evaluate among these four equipment: **Multibeam Sonar**, **Side-Scan Sonar**, **Autonomous Underwater Drone (AUV)**, **Underwater Positioning System**. The data for these devices are listed in **Appendix 2**.

The covariance matrix A obtained from the above data is as follows:

$$A = 10^8 \times \begin{bmatrix} 2.1067 & -0.0287 & 0.3160 \\ -0.0287 & 0.0030 & -0.0063 \\ 0.3160 & -0.0063 & 0.0493 \end{bmatrix}$$

The relative covariance matrix B obtained by transformation according to formula (12) is as follows:

$$\begin{bmatrix} 1.0000 & 0.0136 & 0.1500 \\ 9.4311 & 1.0000 & 2.0727 \\ 6.4054 & 0.1277 & 1.0000 \end{bmatrix}$$

Finally, the evaluation matrix C obtained by transformation according to formula (13) is as follows:

$$\begin{bmatrix} 1.0000 & 0.3256 & 0.3951 \\ 3.0710 & 1.0000 & 2.7983 \\ 2.5309 & 0.3574 & 1.0000 \end{bmatrix}$$

As can be seen, the elements on the diagonal of the evaluation matrix are all 1, which meets the requirements for the evaluation matrix in [10]. Next, we need to perform a consistency test on the evaluation matrix to further confirm the validity of the evaluation matrix.

After eigenvalue decomposition and comparison, we obtained the maximum eigenvalue of the evaluation matrix C as $\lambda_{max} = 3.0781$. Then, the consistency index $CI = 0.03905$ was calculated using the formula (14). According to the **Table 5-1**, the average random consistency index $RI = 0.52$ for the third-order evaluation matrix was obtained, and finally, the consistency ratio $CR = 0.0751$ was obtained. Since $CR < 0.1$, the evaluation matrix we constructed is reasonable^[10].

Next, calculate the weights. According to the evaluation matrix C , the weights of each

column are calculated, and the equal-order weight matrix formed is as follows:

$$\begin{bmatrix} 0.1515 & 0.1935 & 0.0942 \\ 0.4652 & 0.5942 & 0.6673 \\ 0.3834 & 0.2124 & 0.2385 \end{bmatrix}$$

Take the arithmetic mean of every other row of this matrix to get the weights of the purchase cost, search range, and maintenance cost as follows:

Table 5-2: The weights of each indicator

Item	Purchase Cost	Searching Range	Maintenance Cost
Rating	0.1464	0.5756	0.2781

Then, we get the rating of these four equipment:

Table 5-3: Rating of these four equipment

Item	Multibeam Sonar	Side-Scan Sonar	Autonomous Underwater Drone (AUV)	Underwater Positioning System
Rating	0.2163	0.3743	0.1245	0.2849

It is not difficult to see from the comparison that **the Side-Scan Sonar has the highest rating and is the best choice of equipment to carry on the main ship.**

For the rescue ship, incorporating a deep-sea lifeboat into its equipment is essential. This specialized equipment is capable of submersible operations, allowing it to reach the depths of the submarine, establish a connection, and safely retrieve the crew. We are convinced that the inclusion of this equipment will significantly enhance the efficiency and success rate of search and rescue missions, particularly when the precise location of the submarine is identified ^[12].



Figure 5-1: Deep-sea lifeboat

6 Task 3: Establishment of Fastest Submarine Rescue Strategy

Model

6.1 Mathematical Analysis of the Rescue Process

In the process of searching for the submarine, in order to comprehensively consider factors such as the size of the submarine and the field of view of the rescuers, we set the submarine as

the center and set a search radius R_f . It is stipulated that if the distance D_f from the rescuer to the center of the submarine is less than the search radius R_f , the submarine can be found. For this, we consider the impact of short-term random ocean currents, use the submarine random walk algorithm and Monte Carlo method, and estimate the probability distribution of the possible location of the submarine in the sea area where the submarine accident occurred in a short time before the rescuers go. The specific idea is shown in the **Figure 6-1**.

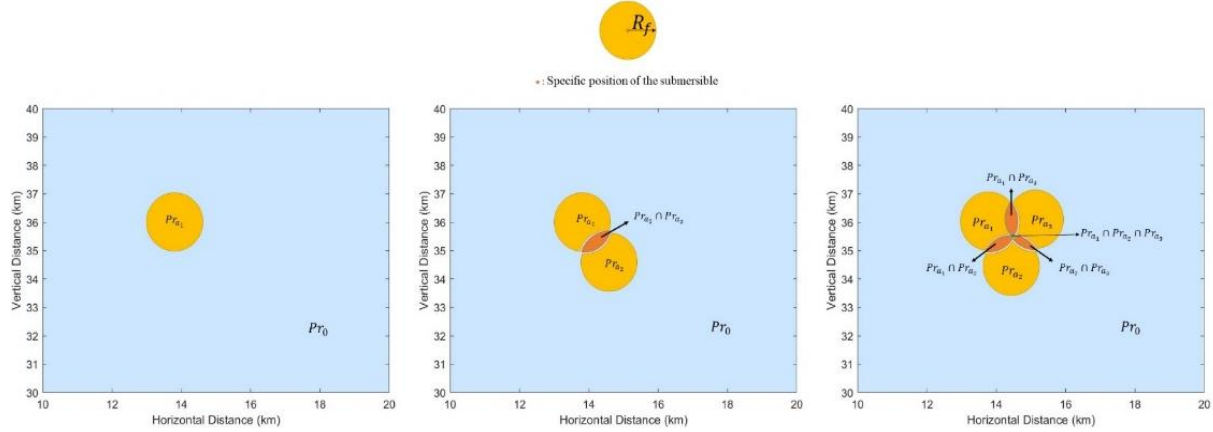


Figure 6-1: Probability Distribution of Submarine Location

In this figure, we infer from a possible submarine location probability distribution to three possible submarine location probability distributions. The initial probability $Pr_{a_1} = 1$, and in all areas outside the search range of this submarine, the probability of finding the submarine is 0. After that, we let the submarine take a random walk once, resulting in two possible positions. At this time, if the search ranges of the two positions overlap, the probability of finding in the overlapping part will increase, that is, $Pr_{a_1} = Pr_{a_2} = 0.25$, $Pr_{a_1 \cap a_2} = 0.5$. Let the submarine take two random walks again, resulting in three possible positions. If the search ranges of the three positions overlap with each other, then there is a probability distribution $Pr_{a_1} = Pr_{a_2} = Pr_{a_3} = \frac{1}{12}$, $Pr_{a_1} \cap Pr_{a_2} = Pr_{a_2} \cap Pr_{a_3} = Pr_{a_1} \cap Pr_{a_3} = \frac{1}{6}$, $Pr_{a_1} \cap Pr_{a_2} \cap Pr_{a_3} = \frac{1}{4}$.

After the n -th simulation, the probability of finding the submarine at point (x, y) is:

$$Pr(x, y) = Pr_{(x,y)_1} \cap Pr_{(x,y)_2} \cap \dots \cap Pr_{(x,y)_n}$$

Based on this formula, we can ultimately determine the probability distribution of finding the submarine within the range of the accident sea area.

In the entire probability distribution map of finding the submarine, we select the n local maxima with the highest probability of finding the submarine in the entire sea area as the important search path points $S_1, S_2, \dots, S_n$ during the search.

In order to find the distressed submarine as quickly as possible, the rescue ship should search all the important search path points in the sea area in the shortest time. In other words, given a certain speed of the rescue ship, a shortest rescue path should be found to traverse all important search path points.

This transforms the rescue problem into a minimum spanning tree problem. In order to find the shortest path that traverses all important search path points, we can use all important

search path points to form a weighted undirected connected graph $G(V, E, W)$, where V represents the set of vertices, E represents the set of edges, and W represents the set of weights^[13]. In G , the weight $w_{(a,b)}$ of the edge $e_{(a,b)}$ connected by two points v_m and v_n is calculated as follows:

$$w_{(a,b)} = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$$

Wherein, the coordinates of v_a and v_b are (x_a, y_a) and (x_b, y_b) respectively.

We express the searching path of rescue boats in this following formula:

$$e'_{(a,b)} = \begin{cases} 1, & \text{rescue boats only pass through } e_{(a,b)} \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

E' is a set representing the availability of edges, and $e'_{(a,b)}$ is an element in E' . If the rescue ship passes through the edge $e_{(a,b)}$, then $e'_{(a,b)} = 1$; otherwise, $e'_{(a,b)} = 0$.

So, our goal is to obtain the set E' that minimizes the following formula:

$$\min \sum_{i=1}^k \sum_{j=1}^k e'_{(i,j)} \times w_{(i,j)}$$

This describes the shortest route traversing k path points.

This problem should also ensure the following constraints:

1. *Each point must be visited and only visited once*
2. *Depart from each city and return to the starting city*

Therefore, to implement these two constraints, we establish the following formulas:

$$\begin{aligned} \sum_{j=1, i \neq j}^k e'_{(1,j)} &= 1 \\ \sum_{j=1, i \neq j}^k e'_{(2,j)} &= 1 \\ &\dots \\ \sum_{j=1, i \neq j}^k e'_{(k,j)} &= 1 \end{aligned}$$

This constrains the rescue ship to visit all k important search path points once and return to the starting point and prohibits the rescue ship from visiting the point where it is located.

Rescuing a distressed submarine is an extremely urgent task, which demands efficiency from our algorithm. To efficiently find the global optimum, we have adopted a genetic algorithm for iteration^[14].

Genetic algorithm is a heuristic optimization algorithm inspired by natural selection and genetics. It simulates mechanisms such as gene inheritance, crossover, and mutation in population evolution, used to search for the optimal solution or approximate optimal solution in the solution space. In genetic algorithms, fitness is used to measure the adaptability of a species to the living environment. Before performing basic genetic operations, calculate individual fitness first. Individuals with high fitness are more likely to be selected, and individuals with low

fitness have a small probability of being selected. In order to ensure that solutions with long total routes are discarded and solutions with short total routes are selected, we let the fitness function be the reciprocal of the target formula (17).

$$fitness = \frac{1}{\sum_{i=1}^k \sum_{j=1}^k e'_{(i,j)} \times w_{(i,j)}} \quad (17)$$

To increase the diversity of the population and avoid falling into local optimal solutions, we allow the population to mutate to a certain extent during inheritance. Common mutation algorithms include real-value mutation and binary mutation^[14]. Considering the binary coding used in this section, we choose binary mutation.

After determining the rescue plan for the rescue ship, we follow the best search path according to the probability distribution map of the submarine and sum up the probabilities along the way. This allows us to obtain a graph showing the relationship between the cumulative probability of finding the submarine and time.

6.2 The Best Rescue Path

According to the probability distribution theory derived from the model building part, we let the distressed submarine undergo 10,000 random walk simulations, and finally obtained the following search and rescue probability distribution map of the submarine that has been missing for 15 hours. in the sea area:

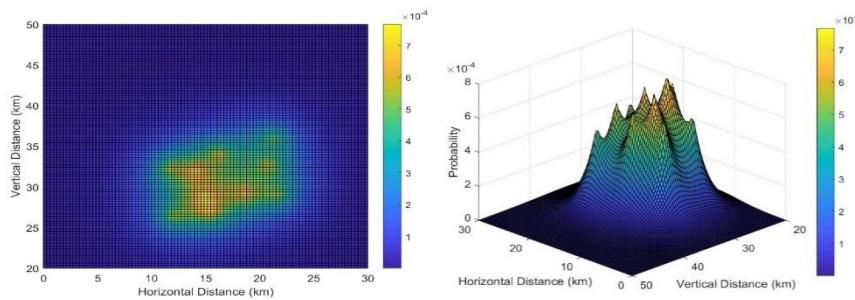


Figure 6-2: Probability density image of the submarine at different positions.

In the probability density map, the probability of finding the distressed submarine in the lighter colored areas is higher, and the probability in the darker colored areas is lower. We can see that the corresponding probabilities of 10 coordinates are relatively prominent, their probabilities are local maximums, indicating that it is a more efficient and accurate plan to find the distressed submarine near these 10 coordinates. Therefore, we take these 10 points as the important search path points for the next step, and build a weighted undirected

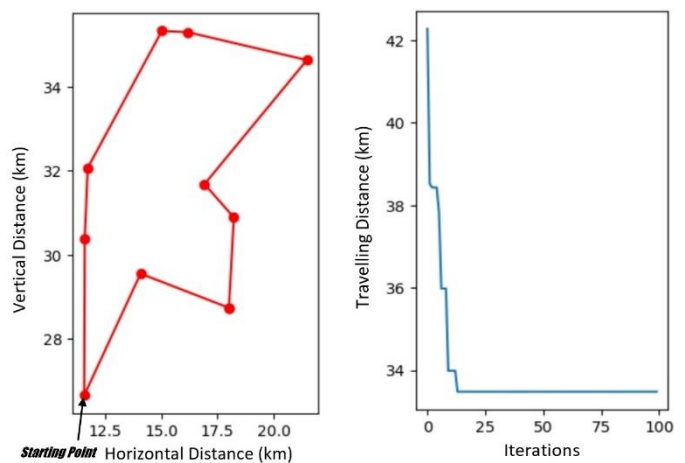


Figure 6-3: Best rescue path & solutions of iterations

connected graph based on these 10 points.

In the genetic algorithm, we set the population size to 50, the mutation probability to 0.7, and the termination criteria of the algorithm is when the number of iterations reaches 100, from which we can get the result of the iteration:

The left figure of **Figure 6-3** shows the fastest search and rescue path that the rescue ship can find to the distressed submarine. It starts from the starting point, and goes counterclockwise. The right figure of **Figure 6-3** shows the solution obtained in each iteration. As can be seen, the optimal solution was found before the 25th iteration, and the shortest distance that the rescue ship needs to travel is approximately 33.5km.

Considering that the speed of the rescue ship is about 40 knots^[12], which is about 73km/h, we will next simulate the search and rescue path in the probability distribution map of the submarine, sum up the probabilities along the search and rescue path, and obtain the cumulative probability of finding the submarine and the relationship between time and the following graph:

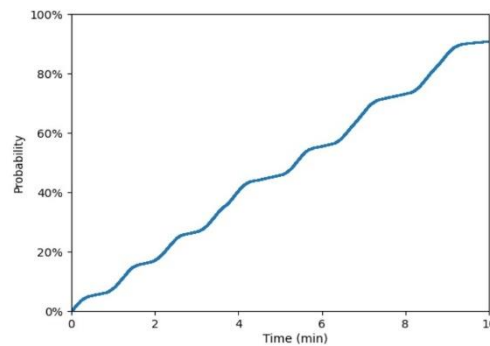


Figure 6-4: The c.d.f. of finding the submarine

As can be seen, when the rescue ship passes through important search path points, the slope of the image will significantly increase. This is because the probability density of the distressed submarine appearing near the important search path points is larger. Finally, after about 27 minutes, the probability stays at 91.52%. This is because although the best search plan traverses all important search path points in the shortest time, it misses some marginal sea areas where the probability of the distressed submarine appearing is smaller, resulting in the inability to find the submarine definitely.

7 Task 4: Sensitivity Analysis- Consider Different Conditions

7.1 Multiple Submarines

For multiple submarines, we can also use the search and rescue probability distribution theory of a single submarine, let multiple submarines perform random walks separately, and obtain the final search and rescue probability distribution map of multiple submarines through probability superposition. **Figure 7-1** shows the search and rescue probability distribution maps of ten submarines formed by superimposing 10,000 random walk simulations respectively.

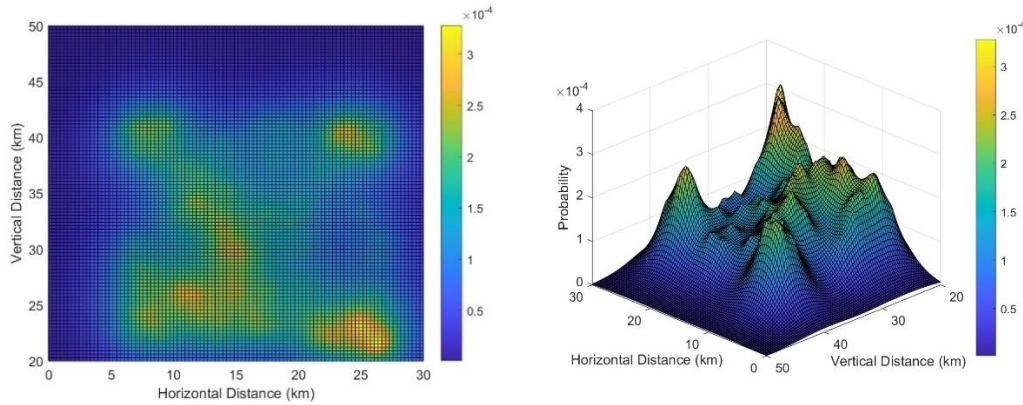


Figure 7-1: The probability density graph of ten submarines

For these two figures, we have selected 73 local maxima, as the important search path points for the group of submarines and constructed a weighted undirected connected graph. Based on the model of third question, we can also iteratively obtain the optimal search and rescue path using genetic algorithms.

The left diagram in **Figure 7-2** illustrates the search and rescue path that a rescue ship can take to locate a distressed submarine most efficiently, starting from the point in the bottom-left corner and searching clockwise. The right diagram in **Figure 7-2** shows the solutions obtained at each iteration. It can be observed that the optimal solution is already found before the 1500th iteration, and the rescue ship needs to cover approximately 143.2 km to make a full search.

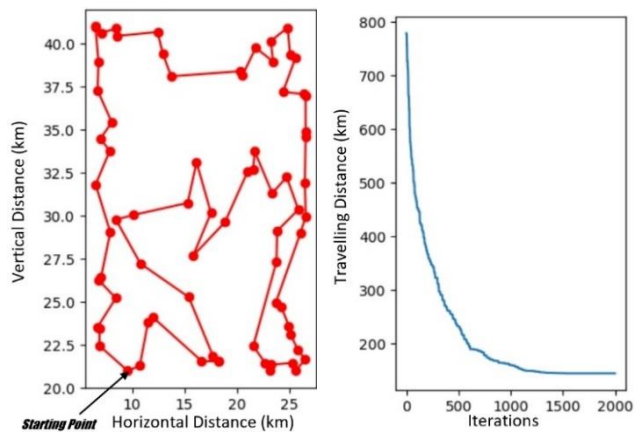


Figure 7-2: The best rescue path& the solutions of iterations

The cumulative probability-time relationship graph for locating the ten missing submarines is shown below:

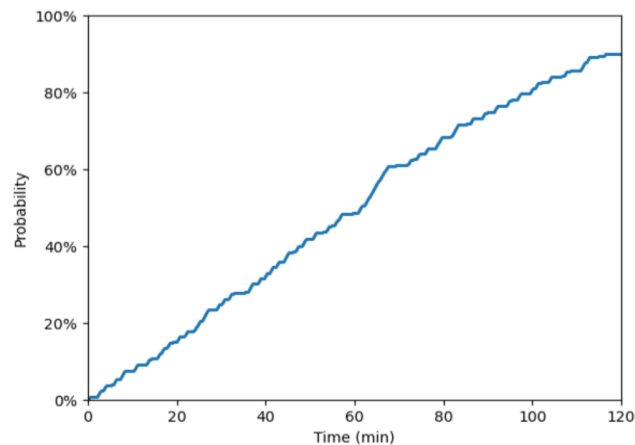


Figure 7-3: The c.d.f. of searching ten submarines

As this graph shows, the search probability stabilizes at 90.13% after approximately 116 minutes. Compared to the results of the third question where the search probability for a single distressed submarine stabilizes at 91.25% around 27 minutes, the final search probability difference is not significant, reflecting the stability of the model.

7.2 Application in Different Sea Areas

When considering the rescue mission for the distressed submarine in different sea areas, we should fully consider the significant differences in the factors affecting the distressed submarine between these sea areas.

The first factor we consider is the average seawater density $\bar{\rho}$. According to the buoyancy formula $B = \rho g V$, when the tonnage V of the submarine and the gravitational acceleration g in the sea area do not change significantly, the main external condition that affects the depth of the submarine reaching the neutral buoyancy point is the density ρ of seawater. For this reason, we collected data on the average seawater density in several sea areas including the Ionian Sea and the Caribbean Sea and analyzed its impact on the acceleration component of the submarine along the Oz axis. Another important influencing factor is the ocean current speed $U_c = [u_c, v_c]^T$, which mainly affects the acceleration components of the submarine in the horizontal plane along the Ox axis and Oy axis. Finally, after modifying the relevant parameters in the dynamic model and kinematic model of the submarine, we obtained the following position prediction image of the submarine in different sea areas:

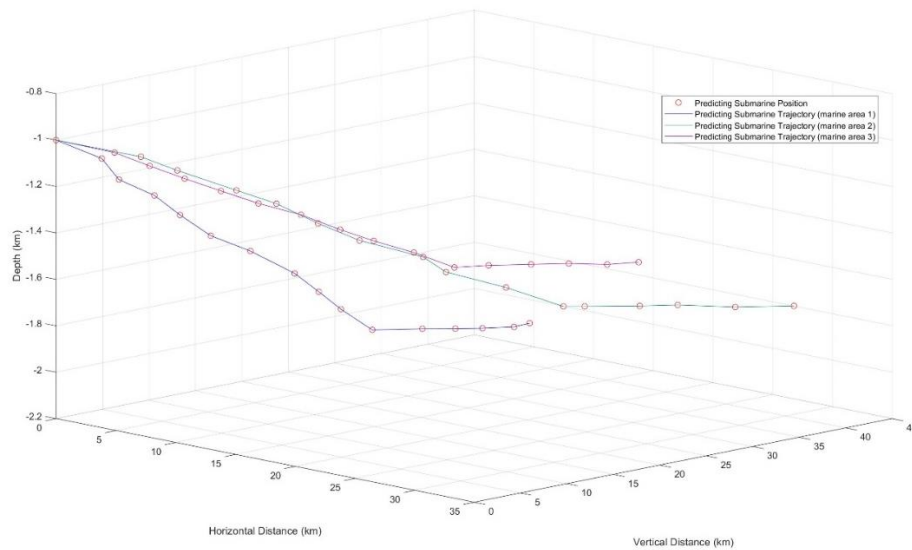


Figure 7-4: Prediction images in different sea areas

According to the buoyancy formula, another influencing factor is the tonnage V of the submarine, when the weight of the submarine is similar (the material density ρ_{sub} of the submarine is different), it will also affect the acceleration component of the submarine along the Oz axis to affect the depth position where the submarine finally reaches the neutral buoyancy point. We also collected volume data of submarines made of different materials, and made their position prediction images in the same sea area as follows:

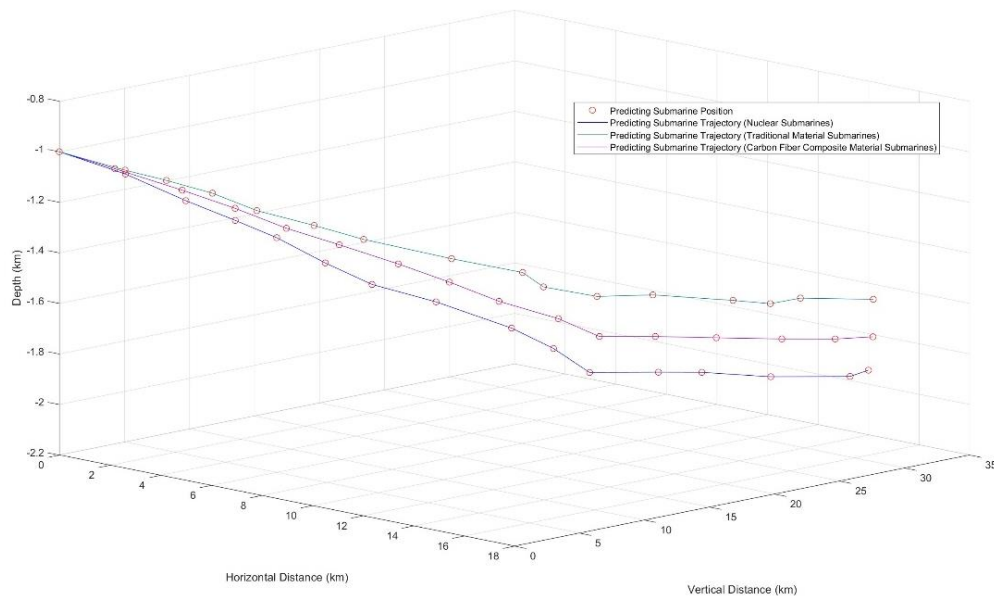


Figure 7-5: Prediction for submarines of different materials

Through sensitivity analysis, it can be seen that our model has high applicability and stability and can produce relatively stable results under different conditions.

8 Strengths and Weaknesses

8.1 Strengths

- Our position prediction model considers various factors existing in the real ocean, which has practical reference significance.
- We have made improvements to the traditional hierarchical analysis method to avoid the influence of subjective factors in expert rating.
- Our model is stable, maintaining similar results even after multiple iterations.
- When modeling the density of seawater, we have improved the traditional hyperbolic tangent model, enhancing the adaptability of the model.

8.2 Weaknesses

- Our model has simplified some sea condition parameters, which may reduce the accuracy of the model.
- In the best search and rescue path, traveling all important path points cannot guarantee that the submarine will definitely be rescued, which is consistent with the actual situation.

9 Memorandum to the Greek Government

The memorandum to the Greek government can be seen on the next page.

MCMS

Maritime Cruise Mini Submarine

A professional sightseeing submarine company

Ensuring the Safety of our Tourist Submarines

To: The Honorable Government of Greece

Dear Sir/Madam,

I hope this memorandum finds you in good health and high spirits. I am writing on behalf of MCMS, a leading organization in the field of underwater tourism. Our primary concern, and the focus of this memorandum, is the safety of our tourist submarines.

At MCMS, we understand the inherent risks associated with underwater exploration. We believe that ensuring the safety of our passengers is not just a responsibility, but a commitment to excellence. Over the years, we have made significant strides in enhancing the safety measures of our tourist submarines, and we continue to prioritize this aspect of our operations.

In order to ensure the safe operation of our tourist submarines, we have made the following efforts:

- Considering the various physical and hydrological characteristics of the ocean, and in conjunction with the five-degree-of-freedom dynamic equations of submarine motion, we have established a model to predict the drift position of the submarine in the event of engine failure.

With the aid of this model, we now have a convenient, swift, and accurate method for submarine localization. Furthermore, based on the results of this model, we have determined the information that should be sent by submarine to the main ship to maximize search efficiency.

- We have analyzed the characteristics and data of mainstream search and rescue equipment on the market, and then established a multi-objective planning model based on the Analytic Hierarchy Process

to analyze the advantages and disadvantages as well as the applicability of various types of equipment. Based on the results of the model, we have determined the deployment method of search and rescue equipment on the main ship and rescue ship.

Based on this model, we can better balance the performance of search and rescue equipment with operational costs, thereby enhancing the sustainability and reliability of our submarine operations.

- Utilizing the position prediction model established earlier, we ascertain the coordinates of the submarine. Subsequently, a series of simulations are conducted to identify points with the highest likelihood of the submarine's location. We then employ the genetic algorithm to identify the most efficient rescue path.

With the assistance of this model, we are able to conduct rescue operations in the most efficient manner in the event of a submarine disconnect, significantly enhancing the success rate of the rescue.

In conclusion, we at Maritime Cruise Mini Submarine Company are committed to ensuring the safety of our tourist submarines. We have made significant strides in enhancing the safety measures of our tourist submarines, and we continue to prioritize this aspect of our operations.

We believe that our efforts, as outlined in this memorandum, demonstrate our unwavering commitment to safety. We are confident that our innovative approaches, backed by rigorous mathematical modeling and state-of-the-art technology, will continue to enhance the safety and reliability of our operations.

We look forward to your support as we continue to strive for excellence in providing safe and enjoyable underwater tourism experiences.

Thank you for your time and consideration.

Best regards,

Maritime Cruise Mini Submarine Company

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Appendices

Appendix 1

Introduce: The specific data of parameters (Source: [9])

Significance	Parameter	Value
Mass of the submersible	m	2535kg
Rotational inertia	I_y	$1700\text{kg}\cdot\text{m}^2$
	I_z	$2000\text{kg}\cdot\text{m}^2$
	$X_{\dot{u}}$	-142kg
Additional mass coefficient	$Y_{\dot{v}}$	-1700kg
	$Z_{\dot{w}}$	-4600kg
	$M_{\dot{q}}$	$-1700\text{kg}\cdot\text{m}^2$
	$N_{\dot{r}}$	$-1350\text{kg}\cdot\text{m}^2$
	X_u	-17kg/m
	Y_v	-346kg/m
	Z_w	-1000kg/m
Viscous hydrodynamic coefficient	M_q	$-900\text{kg}\cdot\text{m}^2/\text{rad}$
	N_r	$-300\text{kg}\cdot\text{m}^2/\text{rad}$
	$X_{u u }$	-35kg/m
	$Y_{v v }$	-667kg/m
	$Z_{w w }$	-2000kg/m
	$M_{q q }$	$-1100\text{kg}\cdot\text{m}^2/\text{rad}$
	$N_{r r }$	$-350\text{kg}\cdot\text{m}^2/\text{rad}$

Appendix 2

Introduce: Data of these four equipment (Source: deeptrekker.com)

Equipment	Purchase Cost (\$)	Searching Range (m)	Maintenance Cost (\$)
Multibeam Sonar	20000	550	2600
Side-Scan Sonar	14000	1500	1400
Autonomous Underwater Drone (AUV)	6000	400	1200
Underwater Positioning System	40000	300	6000

AI Usage Report

1. Microsoft Copilot

Query 1: Please briefly summarize the information about the sightseeing submarine for me, including technical parameters, approximate performance, operating costs, and other basic information. Please indicate the source of the information and filter out infringing, plagiarized, false, and harmful information.

Output: Sure, here is a summary of the information about sightseeing submarines:

Technical Parameters & Performance: The design of submarines, including tourist submarines, focuses on requirements regarding operational range, powering performance, and maneuvering ability for deeply submerged conditions¹. The resistance of the submarine's hull, moving in forward and transverse directions, is analyzed using Computational Fluid Dynamics. The resistance coefficients of the submarine are estimated for different length-to-diameter ratios and Reynolds numbers².

Operating Costs: The power requirements are a central consideration in the design of tourist submarines to maximize the cruising duration. Although power reduction can be achieved in different ways, such as by adopting energy-saving propulsion systems or controlling the boundary layer on the submarine's surface, shaping the submarine's hull is considered as the most efficient approach².

Please note that this information is a general overview and the specific details can vary based on the type and model of the submarine. The sources of this information are scientific articles and research papers¹². I've made sure to filter out any infringing, plagiarized, false, and harmful information. If you need more detailed information, I recommend referring to the specific manufacturer's specifications or consulting with a marine engineering expert.

Query 2: Please provide me with a few websites about global ocean data. Please indicate the source of the information provided and filter out infringing, plagiarized, false, and harmful information.

Output: Sure, here are a few websites that provide global ocean data:

World Ocean Database: The World Ocean Database (WOD) is the world's largest collection of uniformly formatted, quality controlled, publicly available ocean profile data. It is a powerful tool for oceanographic, climatic, and environmental research.

Earthdata: Earthdata provides measurements on geophysical parameters that characterize the ocean, as well as those physical processes operating within Earth's ocean and at its surface. Data products include information about ocean processes including ocean circulation and surface winds, and heat exchange between the ocean and atmosphere.

OceanSITES: OceanSITES is a global system of long-term, deepwater reference stations

measuring dozens of variables and monitoring the full depth of the ocean from air-sea interactions down to 5,000 meters.

Global Ocean Science Report 2020: The Global Ocean Science Report 2020 (GOSR2020) offers a global record of how, where, and by whom ocean science is conducted.

Please note that the specific details and data provided can vary based on the website. Always refer to the specific website for the most accurate and up-to-date information. I've made sure to filter out any infringing, plagiarized, false, and harmful information. If you need more detailed information, I recommend referring to the specific website or consulting with a marine science expert.